



PLANETARY HEALTH



Institut Pengajian Siswazah



Using modern forecasting models to improve disaster preparedness

Can We Predict Medicine Demand During *FLOODS*?

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WHY IS THIS IMPORTANT?

When floods hit, medicine supply is *at risk*.

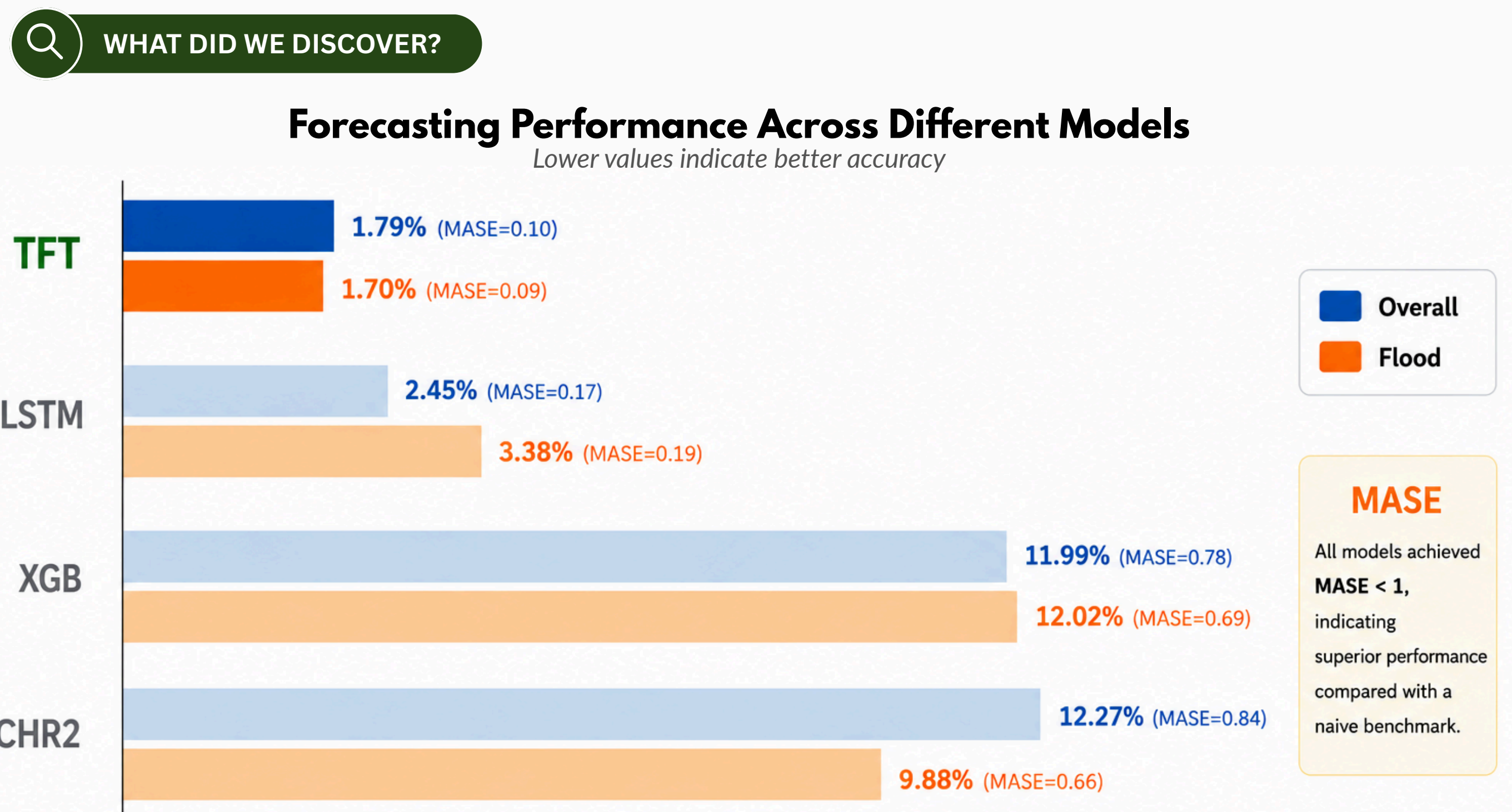
- Roads closed, deliveries delayed, and facilities cut off.
- Demand for heart medicines can rise suddenly.
- Shortages mean treatment delays and health risks.

Can we predict medicine demand before shortages occur?

PREDICT → **PREPARE** → **PROTECT** → **BE READY**

Use forecasting models and real-world data to see what's coming. Plan the right stock, in the right place, at the right time. Reduce shortages and keep patients safe during floods. Stronger preparedness today for a healthier community tomorrow.

Better predictions today. Better care tomorrow. That's disaster-resilient healthcare.



HOW WAS THIS STUDY DONE?

1. DATA & SCOPE

Data from 2017-2023
24 hospitals & clinics in flood-prone areas across Malaysia.

Cardiovascular medicines
3 commonly used medications.

Historical records
Medicine dispensing, population, flood events, weather, and more.

WHAT DATA DID WE USE?

Dispensing Records, Population Data, Weather Data, Flood Data, Operational Data

2. HOW WE DID IT

1. Collect Data → 2. Prepare Data → 3. Build Models → 4. Train & Validate → 5. Compare Models

3. THE FOUR FORECASTING MODELS WE COMPARED

TFT (Temporal Fusion Transformer): Captures complex temporal patterns and uses all relevant information together.

LSTM (Long Short-Term Memory): Good at learning sequential patterns and long-term dependencies.

XGB (eXtreme Gradient Boosting): Powerful tree-based model that handles non-linear relationships well.

CHR2 (Chronos-2): Pre-trained foundation model for time series, adapted to our healthcare data.

STUDY AT A GLANCE

24 Hospitals & Clinics, 3 Cardiovascular Medicines, 7 Years of Data (2017-2023), Flood-Prone Areas, Millions of Records

sMAPE (%) Measures forecasting error in percentage.

MASE Compares against naive forecast.

Key Takeaway: The Temporal Fusion Transformer (TFT) achieved the lowest forecasting error and remained stable during flood periods.

sMAPE: Symmetric Mean Absolute Percentage Error | MASE: Mean Absolute Scaled Error

Information Used by Each Forecasting Model

Information Domain	TFT	LSTM	XGB	CHR2
Temporal Patterns • History • Seasonality	High	High	High	High
Geographic Information • State • District	High	None	Low	None
Disaster-related Features • Flood • Rainfall • Water Level	High	Low	Low	Medium
Operational Context • OPD Visits • Ward Supply • Facility Load	Medium	Medium	None	Medium

KEY TAKEAWAY

TFT leveraged exogenous variables optimally, helping it maintain high forecasting accuracy during flood events.

● High Reliance ● Medium Reliance ○ Low Reliance — No Reliance

WHY DOES IT MATTER?

1. SEE WHAT'S COMING → 2. PLAN THE RIGHT STOCK → 3. RESPOND FASTER → 4. PROTECT PATIENTS → 5. STRENGTHEN HEALTHCARE RESILIENCE

Early insights, smarter decisions. Less guesswork, less waste. Fewer delays, better response. Better outcomes, greater safety. Resilient today, ready for tomorrow.

The Outcome

A Resilient Healthcare Supply Chain

Ensuring the right medicines, at the right place, at the right time — when it matters most.

Right Medicines → Right Place → Right Time → Better Health for All

EXPLORE MORE

Scan to explore our project interactive dashboard, mini game, and more!

<https://radziaziz.github.io/med-flood/>

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